HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python** data pipelines on **AWS Lambda** for ingesting and transforming **~3,500 events/sec** from diverse sources into a centralized data lake.
- Engineered a **Java Spring Boot** microservice to process and validate incoming data streams, ensuring data integrity and adherence to schema requirements for downstream AI model training.
- Designed and implemented **SQL** queries and stored procedures in **PostgreSQL** to extract, transform, and load data for analytical reporting and model feature generation, managing schema evolution across **5** distinct datasets.
- Collaborated with **AI scientists** to understand data requirements for model training, translating complex business needs into technical specifications for scalable data pipelines.
- Optimized data pipeline throughput and reduced processing costs by **18%** through efficient resource management on **AWS EC2** instances and tuning data processing logic.
- Implemented **CI/CD** pipelines using **Jenkins** and **Git** for automated testing and deployment of data pipeline components, reducing deployment time by **40%**.
- Containerized data pipeline services using **Docker** and orchestrated deployments on a **Kubernetes** cluster, ensuring consistent environments and simplified scaling.

SKILLS

Languages – Python, Java, SQL, Bash, R

Cloud Platforms – AWS (EC2, S3, Lambda, CloudWatch)

Databases – PostgreSQL, MySQL, Oracle

Big Data & AI – Data Pipelines, ETL, ML Model Training, Feature Engineering, Data Quality, AI Science Collaboration

DevOps & Tools – Docker, Kubernetes, Jenkins, Git, Jira, Confluence, Splunk, Dynatrace

PROJECTS

Scalable Data Ingestion Pipeline for AI Model Training

- Engineered a robust **Python** data ingestion pipeline to process and aggregate data from multiple enterprise sources, supporting a **~3,400 events/sec** throughput for a fraud detection ML model.
- Integrated with **AWS S3** and **CloudWatch** for scalable storage and real-time monitoring, ensuring high availability and data integrity for critical AI workloads.
- Developed data validation scripts to enforce schema consistency and data quality checks, reducing downstream model training errors by an estimated **22%**.

Real-time Feature Engineering Service

- Built a **Java Spring Boot** service that consumed real-time transaction data, performing feature extraction for downstream AI model inference.
- Implemented asynchronous processing using message queues to handle high volumes of data with p95 latency below **200ms**.
- Containerized the service with **Docker** and deployed to a **Kubernetes** cluster, enabling seamless scaling and management within the production environment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)